📘 Mini-Project Specification: Text Generation with NLTK (Markov Chain based)

Title:

"Generating Shakespeare-style Sentences using NLTK"

Objective:

To demonstrate how Natural Language Processing techniques can be applied to **generate new text** based on an existing corpus, using tokenization and Markov chain–like probabilistic modeling.

Learning Outcomes:

By the end of this project, students will:

- Understand **text preprocessing** using NLTK (tokenization, lowercasing, punctuation removal).
- Build a **bigram language model** (word-to-word probabilities).
- Generate new sentences that **mimic the writing style** of the input text.
- Gain intuition about **statistical language models** (precursor to modern LLMs).

Dataset:

Use any freely available **Shakespeare text** (NLTK comes with one). Example: gutenberg corpus in NLTK (shakespeare-macbeth.txt).

Steps:

1. Load the dataset (Shakespeare text).
2. Preprocess: tokenize, clean, and prepare the corpus.
3. Build a bigram (or trigram) probability model.
4. Implement a simple text generator using random sampling from the probability distribution.
5. Generate and print some sentences to showcase results.

Discuss:

Discuss **limitations** of bigram models:

- Lack of long-term coherence.
- Limited creativity compared to LLMs.

Extend: try **trigrams** instead of bigrams (trigrams(tokens)).

Bonus: to build a **Markov Chain text generator**.